MILITARY SPECIFICATION

LUCKY ASSEMBLY, FOR CARTRIDGES 90MM, HEAT, M371E1 AND 106MM, HEAT, M344A1 PARTS FOR, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specifications covers the parts for, assembling and packing of the lucky assembly for use in Cartridges, 90MM, HEAT, M371E1 and 106MM, HEAT, M344A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-P-10025 - Packaging, Packing and Marking for Interplant Shipment
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105D - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Sampling by Attributes.

FSC 1315

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DRAWINGS

ORDNANCE CORPS

8861654 - Lucky Assembly

PUBLICATIONS

ORDNANCE CORPS

IEL-9200172 - Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assemblies.-The assemblies shall comply with all requirements specified on Drawing (dwg.) 8861654 and with all requirements specified in applicable specifications.

3.3 Activity.-The lucky assembly shall indicate activity when tested in accordance with 4.3.3.1.

3.4 Workmanship.-All parts shall be of the proper finish, dry and free of burrs, cracks, chips, dirt, grease and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier

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may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection

4.2.1 Submission.-Prior to normal production the contractor shall submit an initial production sample consisting of twenty-five consecutively produced assemblies and complete sets of parts (i.e. 25 each of every individual component and every subassembly), which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government approved facility designated by the contracting officer. All parts and materials shall be obtained from the same source of supply as will be used in regular production. The initial production sample shall be accompanied by certificates of analyses for all materials. A sample shall also be submitted whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Each assembly and component shall be subjected by the Government to any or all of the examinations specified in 4.3 of this specification and for all additional requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any or all of the applicable requirements, the initial production sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit new or supplemental quantities until such time as an acceptable sample is submitted, or until otherwise directed by the contracting officer.

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4.3 Inspection provisions.

4.3.1 Lot formation.-A lot shall consist of lucky assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot interfix number. Product shall be submitted for inspection in accordance with Standard MIL-STD-105D (or Standard MIL-STD-1235, when applicable). Each lot shall contain lucky crystals of one lot interfix number from one manufacturer.

4.3.2 Examination.-Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with Standard MIL-STD-105D except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.40 percent for each minor defect and an AQL of 0.25 percent for each major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.3.

4.3.2.1 Plug (see dwg. 8861653 covering a detail of dwg. 8861654).

Categories	Defects	Method of Inspection	Inspection Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Diameter of plug	Gage	01001
Minor:	AQL 2.50 percent		
201.	Length	Gage	01002
202.	Large diameter of head	Gage	01003

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203.	Small diameter of head	Gage	01004
204.	Thickness at large diameter	Gage	01005
205.	Overall thickness of head	Gage	01006
206.	Plug tip radius missing	Visual	01007
207.	Evidence of poor workmanship (see 3.4)	Visual	01008

4.3.2.2 Insulator (see dwg. 7549089 covering a detail of dwg. 8861654).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 2.50 percent		
201.	Outside diameter	Gage	02001
202.	Diameter of hole	Gage	02002
203.	Diameter of counterbore	Gage	02003
204.	Depth of counterbore	Gage	02004
205.	Thickness	Gage	02005
206.	Evidence of poor workmanship (see 3.4)	Visual	02006

4.3.2.3 Container (see dwg. 7549085 covering a detail of dwg. 8861654).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Diameter, max	Gage	03001
102.	Contact tabs missing	Visual	03002
Minor:	AQL 1.50 percent		
201.	Diameter, min	Gage	03003
202.	Height	Gage	03004
203.	Thickness of material	Gage	03005
204.	Evidence of poor workmanship (see 3.4)	Visual	03006

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4.3.2.4 Lucky and container prior to assembly of insulator and plug (see dwg. 8861654).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Improper polority of lucky in container	Visual	04001
Minor:	None defined		

4.3.2.5 Lucky assembly (see dwg. 8861654).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Crimp inadequate to extent that components are loose in container	Visual-Manual	05001
102.	Thickness of crimped container, max	Gage	05002
Minor:	AQL 1.00 percent		
201.	Overall thickness to tab	Gage	05003
202.	Evidence of poor workmanship (see 3.4)	Visual	05004

4.3.3 Testing

4.3.3.1 Activity test - Major defect - Code No. 05005.-This test shall be conducted 100 percent on the completed assemblies. Any assembly which fails to comply with the applicable requirement shall be classed defective and be removed from the lot. This test shall be performed using equipment specified in 4.3.4.

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4.3.4 Inspection Equipment.-Index of Inspection Equipment Lists Number IEL 9200172 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1.1 Level C.-Packing and marking shall be in accordance with Specification MIL-P-10025.

5.2 Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 Inspection Equipment Lists.-Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection Equipment List codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF - Contractor's design responsibility on Ordnance format; ORDM-608-12.
CDCF - Contractor's design responsibility on contractor format.
OD - Ordnance design.
ODMU - Ordnance design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control Drawing.

6.3.3 Ordnance designs.-Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and non-mandatory.

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When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawings.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 Contractor designs.-contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views

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dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608-12, and, in addition, contain the detail and information specified above.

6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1315-A375

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